Reasoning test

For admission to Class 12

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 11 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 11.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 3 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, on these topics.

Maths

1. Number patterns and series
2. Primes, factors, HCF and LCM
3. Fractions
4. Perimeter and area
5. Simple equations
6. Speed, distance and time
7. Averages
8. Ratio and proportion
9. Percentages, profit and loss, interest
10. Squares and cubes
11. Probability

English

1. Vocabulary
2. Grammar
3. Sentence completion and ordering
4. Reading comprehension

Science

1. Matter and its changes
2. Acids, bases and salts
3. Force, motion and energy
4. Heat, light and sound
5. Electricity and magnetism
6. Life processes and the environment

Reasoning

1. Clocks and calendars
2. Number codes and puzzles
3. Everyday units
4. Word problems
5. Time and work
6. Problems on ages
7. Mixtures
8. Venn diagrams
9. Analogies, odd one out, series and codes
10. Reading an experiment
11. Applying a principle to a new situation

Class 11 syllabus questions

Maths

1. Sets
2. Relations and Functions
3. Trigonometric Functions
4. Complex Numbers and Quadratic Equations
5. Linear Inequalities
6. Permutations and Combinations
7. Binomial Theorem
8. Sequences and Series

Science

1. Units and Measurement
2. Motion in a Straight Line
3. Motion in a Plane
4. Laws of Motion
5. Work, Energy and Power
6. Some Basic Concepts of Chemistry
7. Structure of Atom
8. Classification of Elements and Periodicity in Properties
9. Chemical Bonding and Molecular Structure
10. Redox Reactions

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   Find the next number in the series: 1, 2, 6, 24, 120, ___

   A 240 **B** 600
   C 720 **D** 840

2. ENGLISH · BASICS
   Choose the word most nearly opposite in meaning to "transparent".

   A opaque **B** clear
   C bright **D** thin

3. SCIENCE · BASICS
   When a hanging carpet is beaten with a stick, dust falls out of it. This is mainly because

   A the dust stays put as the carpet moves **B** the stick attracts the dust towards it
   C the dust becomes heavier when it is hit **D** the air pushes the dust inwards

4. MATHS · CLASS 11
   If A = {2, 4, 6} and B = {4, 6, 8, 10}, how many elements are there in A ∪ B?

   Type-in answer: _____

5. ENGLISH · BASICS
   The soldier was praised for his ___ in rescuing the trapped villagers.

   A cowardice **B** courage
   C carelessness **D** confusion

6. SCIENCE · CLASS 11
   A car speeds up uniformly from rest to 20 m/s in 4 seconds. How far does it travel in that time, in metres?

   Type-in answer: _____

7. MATHS · BASICS
   A bicycle bought for ₹4,000 is sold at a loss of 15%. What is the selling price, in rupees?

   Type-in answer: _____

8. ENGLISH · BASICS
   Choose the correct words: By the time we reached the station, the train ___.

   A has left **B** leaves
   C is leaving **D** had left

9. SCIENCE · BASICS
   Why is it dangerous to operate an electric switch with wet hands?

   A Wet hands slip off the switch **B** Water lets current pass into the body
   C Water damages the switch's plastic **D** Wet skin makes the switch heat up fast

10. MATHS · CLASS 11
    How many radians is an angle of 90°?

    A π **B** π/2
    C π/3 **D** 2π

11. ENGLISH · BASICS
Asked about the mistake, the cadet gave a candid answer. As used here, "candid" most nearly means

A honest
B short
C rude
D clever

12. SCIENCE · CLASS 11
What is the mass of 0.5 mol of carbon dioxide? (C = 12, O = 16)

A 11 g
B 22 g
C 44 g
D 88 g

13. MATHS · BASICS
Which of these is a perfect square?

A 1,024
B 1,210
C 1,440
D 1,690

14. ENGLISH · BASICS
Choose the correct word: One of my friends ___ a pilot.

A are
B is
C were
D have been

15. SCIENCE · BASICS
An electric kettle draws a current of 5 A from a 230 V supply. What is its power, in watts?

Type-in answer: _____

16. MATHS · CLASS 11
How many two-letter arrangements, with no letter repeated, can be made from the letters A, B, C and D?

Type-in answer: _____

17. ENGLISH · BASICS
Hand is to glove as foot is to ___

A toe
B sock
C walk
D leg

18. SCIENCE · CLASS 11
Which of these elements, all in Group 1, has the largest atomic radius?

A Lithium
B Sodium
C Potassium
D Rubidium

19. MATHS · BASICS
A town's population of 20,000 grows by 10% every year. What will it be after 2 years?

A 24,000
B 24,200
C 24,400
D 26,000

20. ENGLISH · BASICS
"Book the tickets today; the fares go up next week." This advice assumes that

A fares will never come down again
B the train is already nearly full
C the listener still plans to go
D next week is a public holiday

21. SCIENCE · BASICS
On a humid day, drops of water form on the outside of a glass of iced water. This water comes from

A water seeping out through the glass
B the glass giving off moisture
C water vapour in the air condensing
D ice melting inside the glass

22. MATHS · CLASS 11

What is the 6th term of the geometric progression 5, 10, 20, ...?

A 80 **B** 120

C 160 **D** 320

23. ENGLISH · BASICS

In a code, PEN is written as QFO. How is INK written in the same code?

A HMJ **B** KPM

C JOL **D** JNL

24. SCIENCE · CLASS 11

A 1,000 kg car moving at 20 m/s is brought to rest in 5 seconds. What average braking force acts on it?

A 400 N **B** 2,000 N

C 4,000 N **D** 100,000 N

25. MATHS · BASICS

One card is drawn at random from a well-shuffled pack of 52. What is the probability that it is a king or a heart?

A 1/13 **B** 1/4

C 17/52 **D** 4/13

26. ENGLISH · BASICS

When the new rules were announced, the senior cadets took them with a pinch of salt. This means they

A accepted them gladly **B** did not fully believe them

C complained about them loudly **D** followed them very strictly

27. SCIENCE · BASICS

Lights in a house are wired in parallel, not in series. The main reason is that in parallel

A the house draws a smaller total current **B** each bulb glows dimly and saves energy

C one switch turns every bulb on together **D** each bulb gets full voltage on its own

28. MATHS · CLASS 11

What is the coefficient of x^2 in the expansion of $(1 + x)^5$?

Type-in answer: _____

29. ENGLISH · BASICS

A notice on the mess door reads: "Dinner will be served at 8 pm today instead of 7.30 pm." What does it most likely tell us?

A Dinner is usually at 8 pm **B** Dinner is usually at 7.30 pm

C The cook is unwell today **D** Dinner is cancelled today

30. SCIENCE · CLASS 11

A motor lifts a 50 kg load through 10 m in 5 seconds. Taking $g = 10\ m/s^2$, what power does it deliver, in watts?

Type-in answer: _____

Answers

1	**C. 720**	Each number is the one before multiplied by 2, then 3, then 4, then 5. The next step is × 6, and 120 × 6 = 720.
2	**A. opaque**	Transparent means light passes through so you can see through it; opaque means light does not pass through. "Clear" means the same as transparent.
3	**A. the dust stays put as the carpet moves**	The carpet is suddenly pushed away, but the dust tends to stay at rest (inertia), so it is left behind and falls out.
4	**5**	A ∪ B = {2, 4, 6, 8, 10}, which has 5 elements. Adding 3 + 4 = 7 counts 4 and 6 twice. This is a type-in question.
5	**B. courage**	He was praised, so the missing word must be a good quality shown during a rescue. Only courage fits; the rest would be criticised.
6	**40**	Its average speed is (0 + 20) ÷ 2 = 10 m/s, so in 4 s it covers 40 m. Using the final speed for the whole time gives 80 m.
7	**3400**	15% of ₹4,000 is ₹600, so the bicycle sells for 4,000 − 600 = ₹3,400. This is a type-in question.
8	**D. had left**	Two past events are compared, and the train left first. The earlier of two past actions takes the past perfect: "had left".
9	**B. Water lets current pass into the body**	Ordinary water conducts electricity, so moisture can carry current from the switch into the body and cause a shock.
10	**B. π/2**	180° is π radians, so 90° is half of that: π/2. π is a straight angle and 2π a full turn.
11	**A. honest**	Candid means frank and truthful, saying things openly. A candid answer may also be short or blunt, but neither is the meaning of the word.
12	**B. 22 g**	The molar mass of CO_2 is 12 + 2 × 16 = 44 g/mol, so 0.5 mol is 22 g. 44 g is a whole mole; 88 g multiplies by 2 instead of halving.
13	**A. 1,024**	1,024 = 32 × 32. The others fall between squares: 34^2 = 1,156 and 35^2 = 1,225; 37^2 = 1,369 and 38^2 = 1,444; 41^2 = 1,681 and 42^2 = 1,764.
14	**B. is**	The subject is "one", which is singular, so the verb is "is". "Friends" is plural but only describes which group the one belongs to.
15	**1150**	Power is voltage × current: 230 V × 5 A = 1,150 W. This is a type-in question: the answer is typed as a number.
16	**12**	The first letter has 4 choices and the second 3, so $^4P_2$ = 4 × 3 = 12. Counting AB and BA as one gives 6, the number of selections.
17	**B. sock**	A glove is worn on the hand to cover it; a sock is worn on the foot. A toe is part of the foot and walking is what it does.
18	**D. Rubidium**	Atomic radius increases down a group, as each element adds a new shell of electrons. Rubidium is lowest of the four, so it is largest.

19	**B. 24,200**	After one year: 22,000. The second year's 10% is taken on 22,000, adding 2,200, so 24,200. 24,000 forgets the growth compounds.
20	**C. the listener still plans to go**	Advice to book early only makes sense if the listener is going to travel. The other statements are not needed for the advice to hold.
21	**C. water vapour in the air condensing**	The cold glass cools the air touching it, and the water vapour in that air condenses into drops. Glass does not let water through.
22	**C. 160**	The nth term is $ar^{n-1} = 5 \times 2^5 = 160$. 320 uses 2^6, one doubling too many; 80 stops one too early.
23	**C. JOL**	Each letter moves one step forward: P→Q, E→F, N→O. So I→J, N→O, K→L, giving JOL. HMJ moves each letter back instead.
24	**C. 4,000 N**	Force = change in momentum ÷ time = (1000 × 20) ÷ 5 = 4,000 N. 100,000 N multiplies by the time instead of dividing.
25	**D. 4/13**	There are 4 kings and 13 hearts, but the king of hearts is in both, so 16 cards qualify: 16/52 = 4/13. Counting it twice gives 17/52.
26	**B. did not fully believe them**	To take something with a pinch of salt is to doubt it, not to believe it fully. It says nothing about complaining aloud, and it is the opposite of accepting or strictly following.
27	**D. each bulb gets full voltage on its own**	In parallel every bulb gets the full mains voltage and has its own path, so each can be switched on or off without affecting the others.
28	**10**	The term in x^2 is ${}^5C_2\, x^2 = 10x^2$, so the coefficient is 10. ${}^5C_2 = 5 \times 4 \div 2 = 10$. This is a type-in question.
29	**B. Dinner is usually at 7.30 pm**	"Today instead of 7.30 pm" says 7.30 is the time dinner would normally be served, so the usual time is 7.30. The notice gives no reason, so the cook being unwell is a guess, and dinner is moved, not cancelled.
30	**1000**	Work done = mgh = 50 × 10 × 10 = 5,000 J; power = work ÷ time = 5,000 ÷ 5 = 1,000 W. This is a type-in question.